

Peruvian University of Applied Sciences



Faculty of Engineering
Computer Science Program

PC2

COURSE: Introduction to Deep Learning

PROFESSOR: Jairo Pinedo Taquía

NRC: 18522

GROUP MEMBERS:

Student code	Full name
U202210644	Anaya Vadillo, Nathaly Eliane
U202220990	Figueroa Larragán, Marsi Valeria
U20221E167	Quino Neff, Liam Mikael
U202115571	Rodriguez Matos, Loana Colleen

May 2026

Contents

1. Introduction.....	2
2. Methodology.....	2
2.1 Preprocessing & EDA.....	2
2.2 Architecture Implementation.....	3
2.3 Transfer Learning Experiment.....	3
3. Results.....	4
4. Discussion.....	7
5. Conclusions.....	9
6. References.....	9

1. Introduction

Blood cell classification via microscopy is essential for diagnosing hematological conditions. However, manual analysis is resource-intensive. Convolutional neural networks have proven to be highly effective for medical image classification, yet developing practical models requires understanding how network architecture and reuse of pretrained models affect performance. The Blood Cell Image dataset from Kaggle is a public open-domain resource containing over 12,500 microscopic images classified into four blood cell types. Its direct clinical application makes it ideal for investigating these design questions. This project implements and compares two classical CNN architectures (LeNet-5 (LeCun et al., 1998) and VGG-11 (Simonyan & Zisserman, 2015)) trained from scratch on four blood cell types: Eosinophil, Lymphocyte, Monocyte, and Neutrophil. Each architecture is evaluated with and without Batch Normalization to analyze its effect on training dynamics. Additionally, Transfer Learning strategies using a pretrained ResNet-18 are explored to assess the value of pretrained features in a medical imaging context with limited domain-specific data.

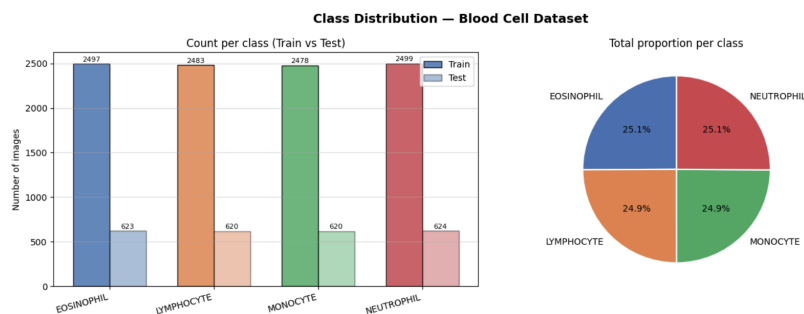
2. Methodology

2.1 Preprocessing & EDA

We conducted exploratory data analysis to understand the dataset's statistical properties and determine necessary preprocessing steps. The dataset contains approximately 12,500 RGB images originally at 320×240 pixels, split into training (~9,957 images) and test (~2,487 images) sets. As shown in Figure 1, the class distribution is remarkably balanced, with each of the four classes: Eosinophil (Train: 2,497, Test: 623), Lymphocyte (Train: 2,483, Test: 620), Monocyte (Train: 2,478, Test: 620), and Neutrophil (Train: 2,499, Test: 624); representing approximately 25% of the total dataset. This balance eliminates the need for class-weighted loss functions or oversampling strategies.

Figure 1

Class distribution of the Blood Cell Image Dataset across training and test splits.



Based on our analysis, images were resized to 64×64 pixels for LeNet-5 and VGG-11, consistent with the original LeNet design philosophy and trainable on CPU/Colab without memory constraints. For Transfer Learning, we used 224×224 pixels to match ResNet-18's expected input size, with ImageNet normalization

statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Data augmentation was applied exclusively to the training split to prevent data leakage, including random horizontal flips, rotations up to 10° , and color jitter (brightness and contrast ± 0.2). The full training set was further divided into 85% training and 15% validation using a fixed random seed of 42 for reproducibility.

2.2 Architecture Implementation

We implemented two classical CNN architectures from scratch, each in two variants: a baseline and one with Batch Normalization (BN), yielding four models total.

LeNet-5 was adapted from LeCun et al. (1998) for RGB input at $64 \times 64 \times 3$ and four output classes. The architecture preserves the original structure: two convolutional-pooling blocks followed by three fully connected layers. The first block applies a 5×5 convolution (6 filters) followed by ReLU and MaxPooling, reducing the spatial dimension from 64×64 to 30×30 . The second block applies a 5×5 convolution (16 filters), further reducing to 13×13 . The flattened output (2,704 neurons) feeds three FC layers (120, 84, 4). ReLU activations replace the original tanh to improve gradient flow. The BN variant inserts a BatchNorm2d layer after each convolution and before ReLU, as recommended by Ioffe & Szegedy (2015).

VGG-11 was adapted from Simonyan & Zisserman (2015) with all filter counts halved to enable training on CPU and Google Colab without memory constraints. The simplified architecture uses five convolutional blocks with filter progression [32, 64, 128, 128, 256, 256, 256, 256], each followed by MaxPooling. An AdaptiveAvgPool layer reduces the output to 2×2 before three FC layers (1,024, 512, 4) with Dropout ($p=0.5$) for regularization. The BN variant inserts BatchNorm2d after each convolutional layer. Total parameters: $\sim 3.1\text{M}$.

All 4 models were trained with the following hyperparameters:

- Optimizer: Adam
- Learning Rate: $1e-3$
- Batch Size: 32
- Epochs: 20
- Random Seed: 42

2.3 Transfer Learning Experiment

We employed ResNet-18 (He et al., 2016) pretrained on ImageNet to investigate three Transfer Learning strategies, each differing in the number of trainable layers:

1. Feature Extraction: Frozen convolutional layers with only the final fully connected layer trainable (Learning Rate of $1e-3$).

2. Partial Fine-tuning: First two blocks frozen. Final blocks and FC layer trainable (Learning Rate of $1e-4$)
3. Complete Fine-tuning: All layers unfrozen with a reduced learning rate (Learning rate of $1e-5$)

In all three strategies, the original FC layer (1,000 classes) is replaced with a new linear layer mapping to 4 classes. The goal is to compare the performance of all three strategies against the best model trained from scratch (Section 2.2) to assess the value of Transfer Learning in a medical context.

3. Results

Table 1 summarizes the performance of all seven models. Among models trained from scratch, VGG-11 Simple achieved the highest test accuracy at 89.06%. LeNet-5 variants showed significantly lower test accuracy despite high validation accuracy, indicating overfitting due to limited model capacity. Among Transfer Learning strategies, Full Fine-tuning achieved the highest test accuracy at 85.32%, though it remained below the best scratch model.

Table 1

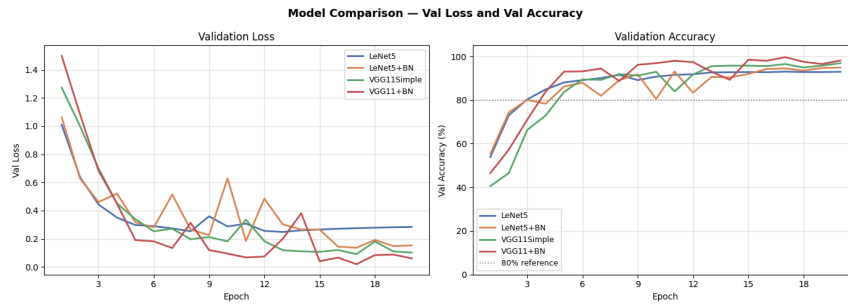
Model Performance and Architecture Comparison Table

Model	Test Acc	Total Params	Trainable Params	Avg. time/epoch	Epochs to 80% Val. Acc.
LeNet-5	63.57%	338K	338K	13.7s	3
LeNet-5 + BN	69.16%	339K	339K	12.3s	3
VGG-11 Simple	89.06%	3.1M	3.1M	12.4s	5
VGG-11 Simple + BN	87.41%	3.1M	3.1M	13.2s	4
ResNet-18 Feature Ext.	60.47%	11.2M	2K	16.9s	-
ResNet-18 Partial FT	80.30%	11.2M	10.5M	16s	2
ResNet-18 Full FT	85.32%	11.2M	11.2M	16.7s	1

The model comparison plot below summarizes the validation loss and accuracy trajectories of the 4 models throughout training.

Figure 2

Model Comparison



LeNet-5 variants reached 80% validation accuracy by epoch 3, while VGG-11 Simple reached it at epoch 5 and VGG-11+BN at epoch 4. LeNet-5 maintains the highest final validation loss, indicating limited model capacity. Both VGG-11 variants continue improving throughout training.

Additionally, training LeNet-5 variants at a higher learning rate ($LR = 0.01$) confirmed that the BN variant maintained stable convergence while the non-BN variant diverged, with validation accuracy dropping to approximately 25% after epoch 5.

To isolate the effect of batch normalization, both architectures were trained under identical conditions with the only variable being the presence or absence of BN layers. For LeNet-5, both variants reached 80% validation accuracy at epoch 3. For VGG-11, BN reduced the epochs needed to reach 80% from epoch 5 to epoch 4, and produced lower and more stable validation loss throughout training.

Figure 3

Effect of Batch Normalization on LeNet-5 Training Dynamics

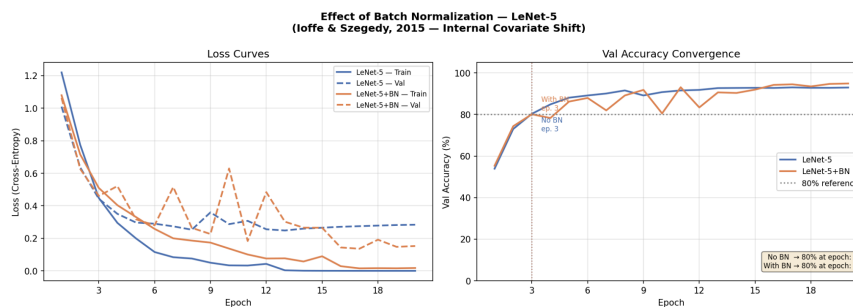
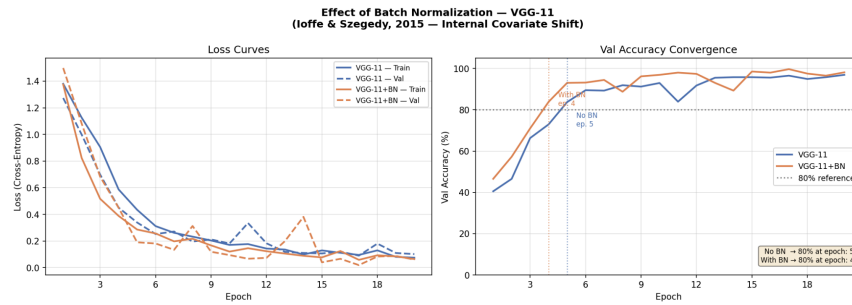


Figure 4

Effect of Batch Normalization on VGG-11 Training Dynamics



Additionally, 3 Transfer Learning strategies were evaluated using a pretrained ResNet-18. Feature extraction achieved only 60.47% test accuracy, never crossing the 80% validation accuracy threshold across all 20 epochs, with validation loss stabilizing around 0.60. Partial fine-tuning achieved near-perfect validation accuracy from epoch 2 onward, reaching a test accuracy of 80.30%. Full fine-tuning showed the smoothest convergence of all models, with both train and validation curves approaching 100% accuracy by epoch 5, achieving the highest TL test accuracy at 85.32%.

Figure 5
Training and Validation Curves for ResNet-18 Feature Extractor

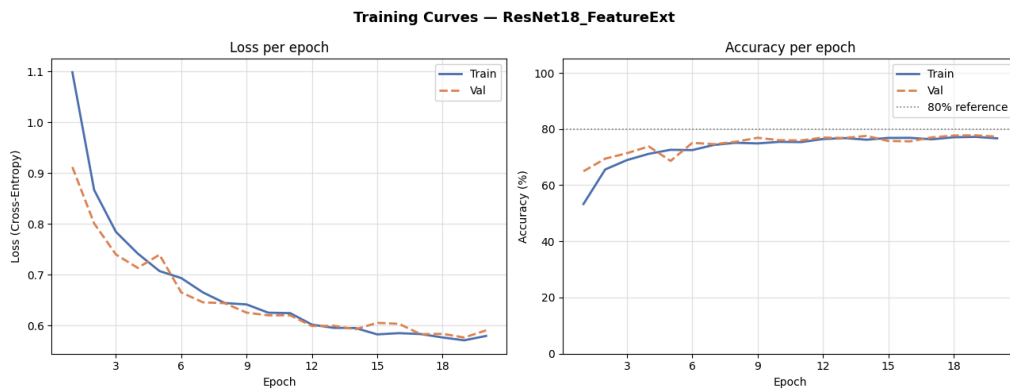
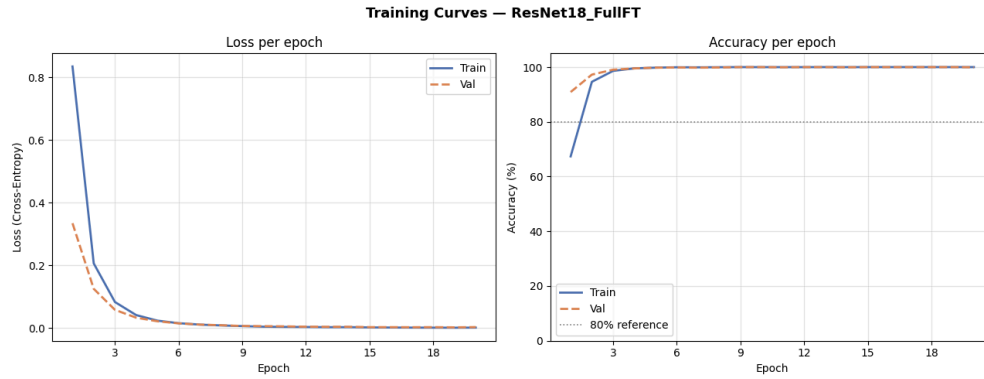


Figure 6
Training and Validation Curves for ResNet-18 Partial Fine-Tuning



Figure 7

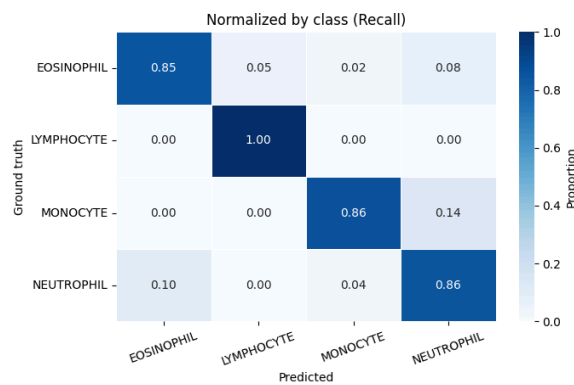
Training and Validation Curves for ResNet-18 Full Fine-Tuning



The confusion matrix for VGG-11 Simple shows balanced per-class recall, with perfect LYMPHOCYTE recall (1.00) and above 0.85 for EOSINOPHIL and NEUTROPHIL. The most frequent misclassification occurred between MONOCYTE and NEUTROPHIL (14%) and between NEUTROPHIL and EOSINOPHIL (10%), which share similar morphological features under microscopy.

Figure 8

Confusion Matrix: VGG-11 Trained from Scratch



4. Discussion

VGG-11 Simple outperformed LeNet-5 in both convergence speed and final test accuracy, consistent with Simonyan & Zisserman (2015), who demonstrated that deeper architectures with smaller convolutional filters achieve stronger feature extraction. LeNet-5, originally designed for grayscale document recognition on low-resolution inputs (LeCun et al., 1998), showed capacity limitations despite the input adaptation, with its training loss reaching near zero while validation loss stabilized around 0.28. This is a sign of overfitting due to the model's limited representational power relative to the task complexity. LeNet-5's architecture was intentionally preserved without additional regularization to maintain the controlled comparison required by the experiment, making this overfitting an expected empirical finding rather than a correctable error.

Batch Normalization, introduced by Ioffe & Szegedy (2015) to reduce internal covariate shift, had notably different effects depending on architecture depth. For LeNet-5, both variants reached 80% validation accuracy at epoch 3, showing no convergence difference at this threshold. For VGG-11, BN reduced the epochs needed to reach 80% from epoch 5 to epoch 4, with the benefit being more pronounced for the deeper model. Regarding loss stability, BN variants produced noticeably smoother validation loss curves with fewer short-term oscillations, consistent with BN's mechanism of normalizing mini-batch activations and stabilizing gradient updates as described by Ioffe & Szegedy (2015). This effect was further confirmed in the high learning rate experiment ($LR = 0.01$), where the BN variant maintained stable training while the non-BN variant exhibited large oscillations and divergence under the same conditions. While BN consistently improved training stability, architecture depth remained the dominant factor since VGG-11 outperformed LeNet-5 regardless of BN, suggesting that capacity and depth cannot be substituted by normalization alone.

The three ResNet-18 Transfer Learning strategies produced different outcomes. ResNet-18 (He et al., 2016) was pretrained on ImageNet and applied to the blood cell dataset using three levels of adaptation. Feature extraction stabilized below 80% validation accuracy across all 20 epochs, with validation loss stabilizing around 0.60. This confirms that frozen ImageNet features lack the domain-specific higher-level representations needed to distinguish blood cell morphologies, a limitation consistent with the observations of Krizhevsky et al. (2012), who noted that deeper convolutional features are more task-specific and therefore more sensitive to domain shift.

Partial fine-tuning achieved near-perfect validation accuracy. Full fine-tuning showed similarly strong performance and has the most stable training behavior of all models tested. Its test accuracy of 85.32% is, however, lower than VGG-11 Simple (89.06%), and only marginally higher than partial fine-tuning (80.30%), despite training significantly more parameters.

Several limitations should be acknowledged. The unexpectedly lower test accuracy of ResNet-18 Full FT compared to VGG-11 Simple+BN warrants caution. The very small learning rate combined with 20 epochs may not have allowed adequate adaptation of early layers. Additionally, validation loss curves for LeNet-5+BN and VGG-11+BN show oscillations across epochs. Feature extraction never reaching 80% accuracy suggests that the visual domain gap between ImageNet and microscopic blood cell images is significant enough to require at least partial fine-tuning.

For future work, increasing the number of training epochs with early stopping based on test set performance could help close the gap between validation and test accuracy observed in the full fine-tuning strategy. For the models made from scratch, stronger data augmentation could reduce the overfitting visible in LeNet-5's train/val loss divergence.

5. Conclusions

VGG-11 Simple outperformed LeNet-5 in both convergence speed and final test accuracy. Furthermore, LeNet-5 training loss fell near zero while validation loss stabilized around 0.28, indicating overfitting and insufficient representational power for the blood-cell classification problem. This behavior was an expected outcome of preserving LeNet-5's original architecture without additional regularization.

Among ResNet-18 transfer strategies, frozen feature extraction failed to exceed 80% validation accuracy, underscoring the domain gap between ImageNet features and microscopic blood-cell morphology. Partial and full fine-tuning produced strong results, with partial fine-tuning achieving the best validation performance and full fine-tuning showing the most stable training but a lower test accuracy than VGG-11 Simple, despite training significantly more parameters. Future work should extend training with early stopping or employing staged unfreezing for ResNet-18. Batch Normalization improved training stability and convergence speed, with its benefits being more pronounced in the deeper VGG-11 architecture. Finally, for medical imaging contexts with limited data, partial fine-tuning is the recommended strategy, as it achieves strong test accuracy with fewer trainable parameters and lower computational cost than full fine-tuning.

6. References

He, K., Zhang, X., Ren, S., & Sun, J. (2016, June 27-30). *Deep residual learning for image recognition* [Conference presentation]. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, USA. <https://doi.org/10.1109/CVPR.2016.90>

Ioffe, S., & Szegedy, C. (2015). Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. *Proceedings of Machine Learning Research*, 37, 448-456. <https://doi.org/10.48550/arXiv.1502.03167>

Krizhevsky, A., Sutskever, I., & Hinton, G. (2012). ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6), 84–90. <https://doi.org/10.1145/3065386>

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324. <https://doi.org/10.1109/5.726791>

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. *Proceedings of the International Conference on Learning Representations (ICLR)*. <https://doi.org/10.48550/arXiv.1409.1556>